Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Missing Data

> Prepare Missing Data: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Detect absent and unusable values

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Noise and Outliers

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Accuracy as a Fraction

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

Concept to explain

A system needs an explicit policy for unknown fields.

Term	Meaning
missing data	Information that was not recorded or is unavailable, which is different from an observed zero.
default	The value or behaviour used when no alternative is specified.
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.
impute	Replace a missing value using a stated method; an imputed value is an estimate, not an observation.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 14 as a two-session investigation using fictional data. A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No. The first is text and the second is an integer.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can validate **missing value**s without confusing zero with absent data.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Model Missing Data using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Reader's length preference is missing.

2. Ask: best when the user can answer.

3. **Abstain**: safest if the field is essential; **default**: convenient but may misrepresent.

4. Record which policy was used so the explanation stays truthful.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Genre is known, but page count is missing and length affects recommendation score.

The teaching case

Choose transparent handling.

Exact explanation to model

Why the result follows

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Explain the added worked example

A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses.

dict.get returns None when this key is absent. An explicit None also represents unknown in this dataset. append adds an item to the end of a list. The script prints [0, 12] and then 3.

The exact type check prevents comparing None or text with the numeric range.

Key terms

Term	Meaning
Missing value	A value that was not recorded
Validation	Checking data against explicit rules
Rejection	Holding a record out because it fails a rule
Coverage	Which intended records or groups are represented

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **Add {"minutes": 60} and {"minutes": "5"}. Predict the accepted list and rejection count.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Missing Data

A fictional catalogue records counts of available copies under a strict **input** contract.

The facts we will use

> **Supplied facts**
>
> Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

> **Supplied facts**
>
> Rows have count 0, count 4, absent count, count "4", count True and count 9.

AI ACADEMY / AI-2/18

Missing Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve carefully

Use row.get("count"); absent field gives None, distinct from 0.



2. Check exact type

Integers 0,4 pass; text "4", Boolean True and None fail.



3. Check range

Integer 9 fails maximum 8.



4. Report validation

Accepted counts [0,4]; four rows rejected with recorded reasons.

CHANGE ONE THING

Replace text "4" with integer 4 through an authorised conversion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve carefully	Use row.get("count"); absent field gives None, distinct from 0.
2. Check exact type	Integers 0,4 pass; text "4", Boolean True and None fail.
3. Check range	Integer 9 fails maximum 8.
4. Report validation	Accepted counts [0,4]; four rows rejected with recorded reasons.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Testing if value would wrongly discard zero, which is a real valid count. Python’s isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

What this example does not establish

Keep the stated limit

Type and range checks cannot confirm whether the real shelf count was measured accurately.

Change one thing in the pictured case

Change one thing

Change: Replace text "4" with integer 4 through an authorised conversion.

Predicted result

Prediction: Accepted values become [0,4,4]; three rows remain rejected.

Why the result follows

Reason: The converted value now satisfies both type and range checks.

Ask while showing the picture

Ask while showing the picture · 1

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Missing Data, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Genre is known, but page count is missing and length affects recommendation score.

2. State the decision to explain

Write the question to explain: Choose transparent handling.

3. Reveal the worked reasoning

Reveal after **prediction**s: Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Hold back the independent task

Keep for an independent attempt: A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant

details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Accepted [0, 12, 60]; 4 rejected. Text "5" fails our rule. Exit: The summary excludes three sessions, which may differ from accepted ones. Report the exclusion and reasons rather than implying full **coverage**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Reasoning to listen for

Teacher check: Accepted [4, 0]; 2 rejected.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Reasoned teacher answer

The contract requires actual integer counts, not Boolean flags; type(True) is bool.

Changed-case reasoning: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Missing Data investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Run the same four incomplete profiles through ask, **abstain**, and **default** policies. Compare usefulness, risk, and user effort.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: integer score 0 through 5 for 5, None, 0, 6 and True. Accept [5, 0]; reject 3. Exact type rejects Boolean True.

A convenient shortcut preserves meaning: Using if value would discard the legitimate zero and accept some invalid nonempty text. Test the written type and range rules explicitly.

Filling an absent duration with 0 falsely claims an observed zero.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Missing Data questions, teacher keys and criteria after the relevant attempts.

> Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Independent question

Independent question · withhold the key

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

Expected independent answer

Teacher key · Expected independent answer

Blank means unknown, not zero. Ask for the page count or mark length unknown and avoid claiming it meets the short-book rule.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: "**Missing data** means the numerical value is zero." Keep the first attempt and an uncoached retry separate.

Original transfer answer

> **Teacher key · Original transfer answer**
>
> Rejecting may lose a good candidate; zero points treats unknown as no match and must be disclosed; asking can resolve uncertainty. A partial score is 2 with length unknown, not a confirmed complete score of 2.
>
> Choose a policy before evaluation.

Explanations for the additional workbook examples

W1 Distinguish zero and unknown

> **Workbook question · W1**
>
> Explain the difference between {"minutes": 0}, {} and {"minutes": None}.

> **Teacher reasoning · W1**
>
> 0 is a recorded allowed duration. The other two are unknown under this schema.

W2 Trace filtering

Workbook question · W2

Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Teacher reasoning · W2

Accepted [4, 0]; 2 rejected.

Teacher reasoning · W2

Process the four entries in order. The integers 4 and 0 have the required type and lie within 0 through 60, so keep [4, 0]. The integer 61 fails the upper boundary.

The text "4" fails the exact integer type, even though its characters look numeric. That makes two rejected entries.

W3 Repair a condition

Workbook question · W3

A program uses if value to accept durations. Give a counterexample and the correct rule.

Teacher reasoning · W3

0 is wrongly rejected; nonempty text can be accepted. Require exact integer type and 0 <= value <= 60.

W4 Validate independently

Workbook question · W4

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

Teacher reasoning · W4

Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries.

W5 Report the limitation

Workbook question · W5

Only 2 of 5 supplied durations were accepted. Why should the report include this?

Teacher reasoning · W5

The summary excludes three sessions, which may differ from accepted ones. Report the exclusion and reasons rather than implying full **coverage**.

Assess independent transfer

Independent question · withhold the key

Selected case: Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

Selected case answer

> **Teacher key · Selected case answer**
>
> Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Missing Data to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 19: Sampling and Representation. Retrieve today’s **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use three piles: accepted zero, accepted other and review. Extension: retain each rejected row with a reason instead of just counting it.

Independent W4 **evidence**: Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries. Record whether the learner needed no help, a hint or explanation.

Fresh retry: integer score 0 through 5 for 5, None, 0, 6 and True. Accept [5, 0]; reject 3. Exact type rejects Boolean True.

Next week: Units conversions and data provenance. Reuse this week's explicit contracts and **evidence**.

Plan the delayed return

Suggested return points in this level: ai-2/19, ai-2/21, ai-2/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Missing Data lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Detect absent and unusable values

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Supported practice answer

Reasoned practice answer: Accepted [4, 0]; 2 rejected.

Check every learner before discussion: Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace text "4" with integer 4 through an authorised conversion. Require a **prediction** and a reason before comparing. Expected result: Accepted values become [0,4,4]; three rows remain rejected.

Reason: The converted value now satisfies both type and range checks.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

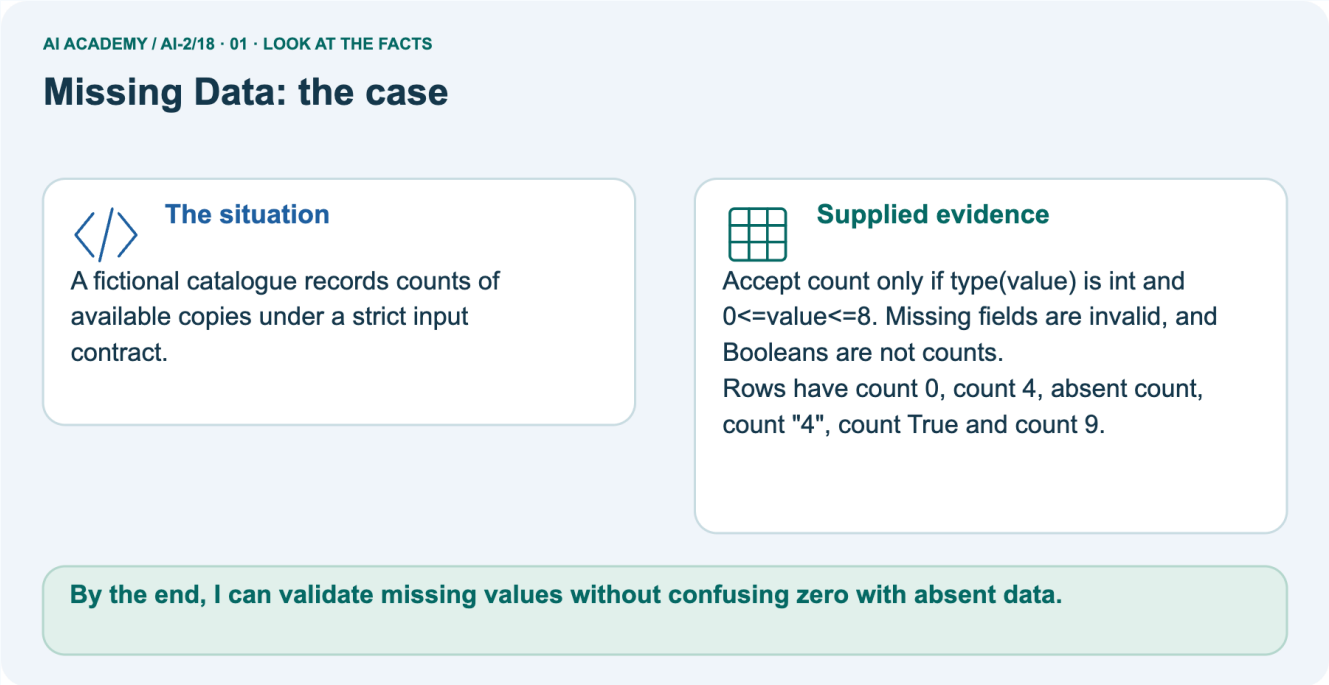
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Missing Data graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Missing Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue records counts of available copies under a strict **input** contract. Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

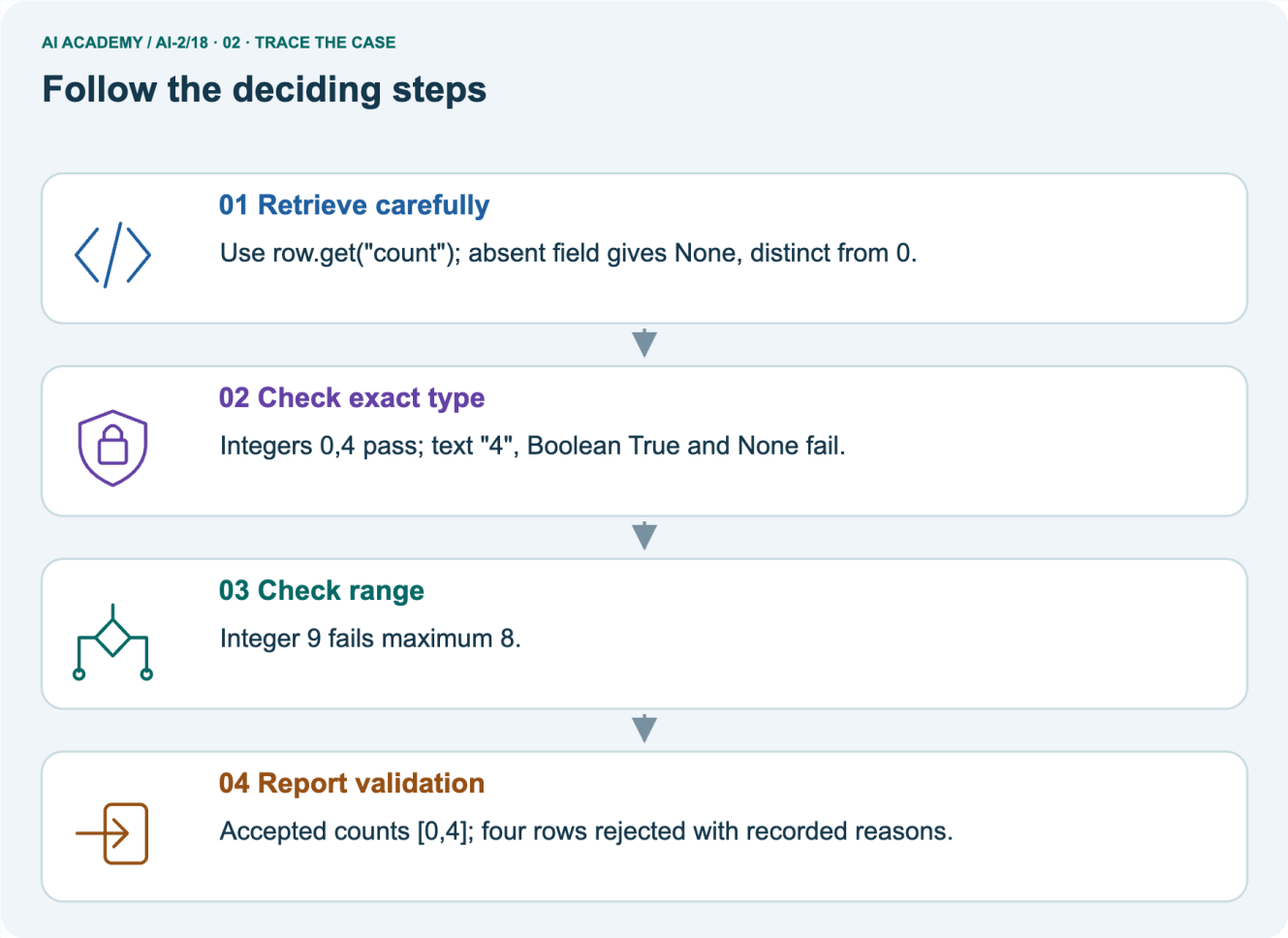
Rows have count 0, count 4, absent count, count "4", count True and count 9.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

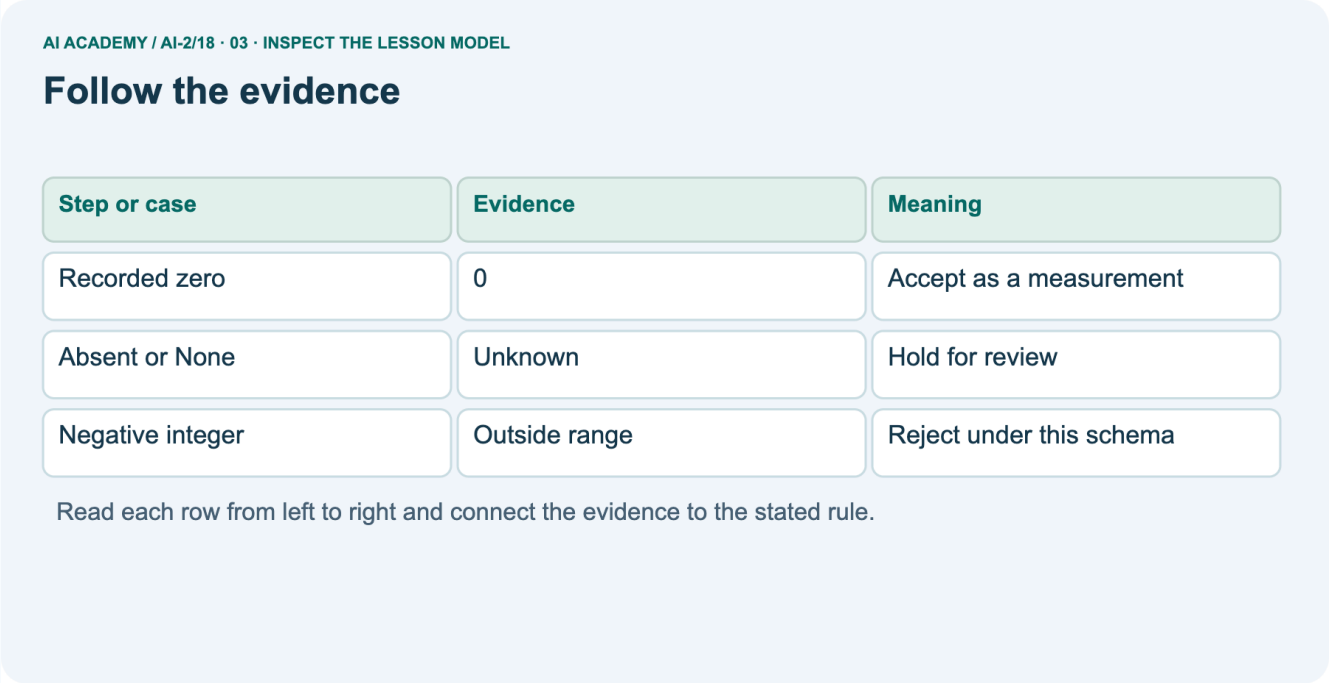
Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

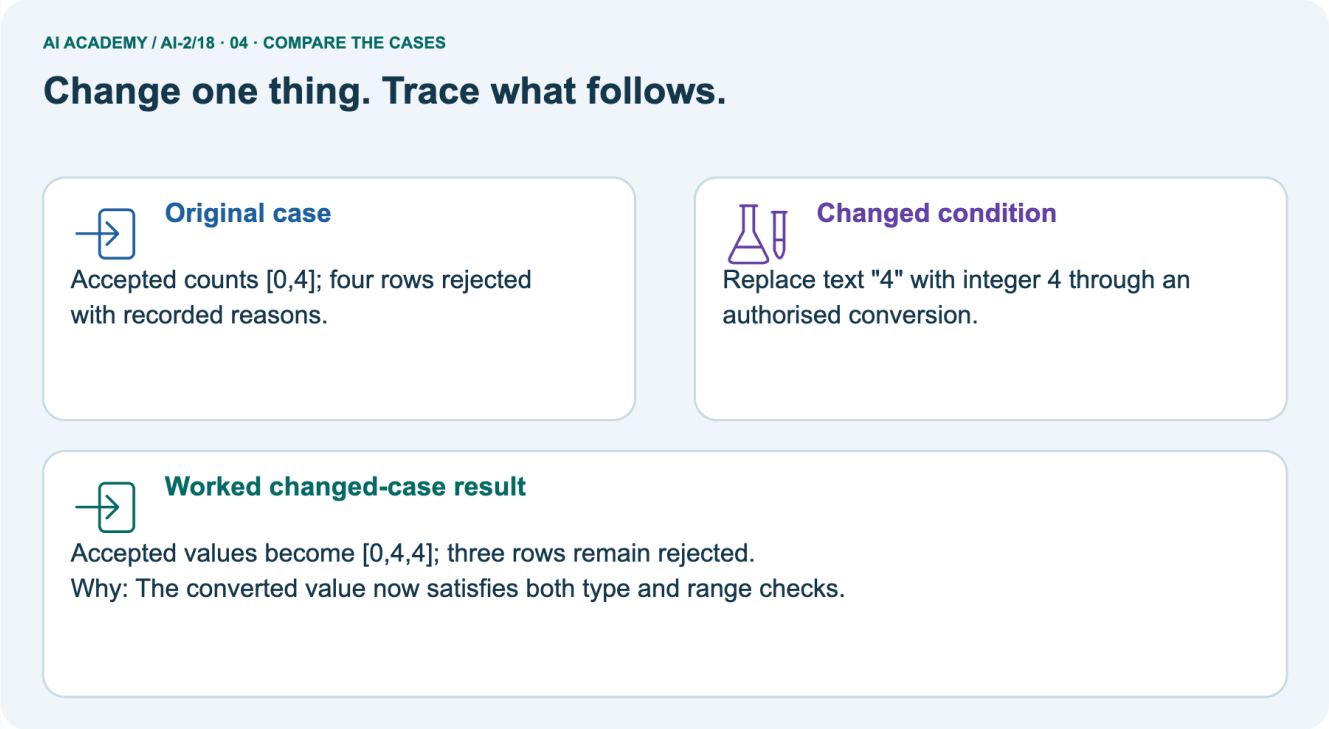
Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

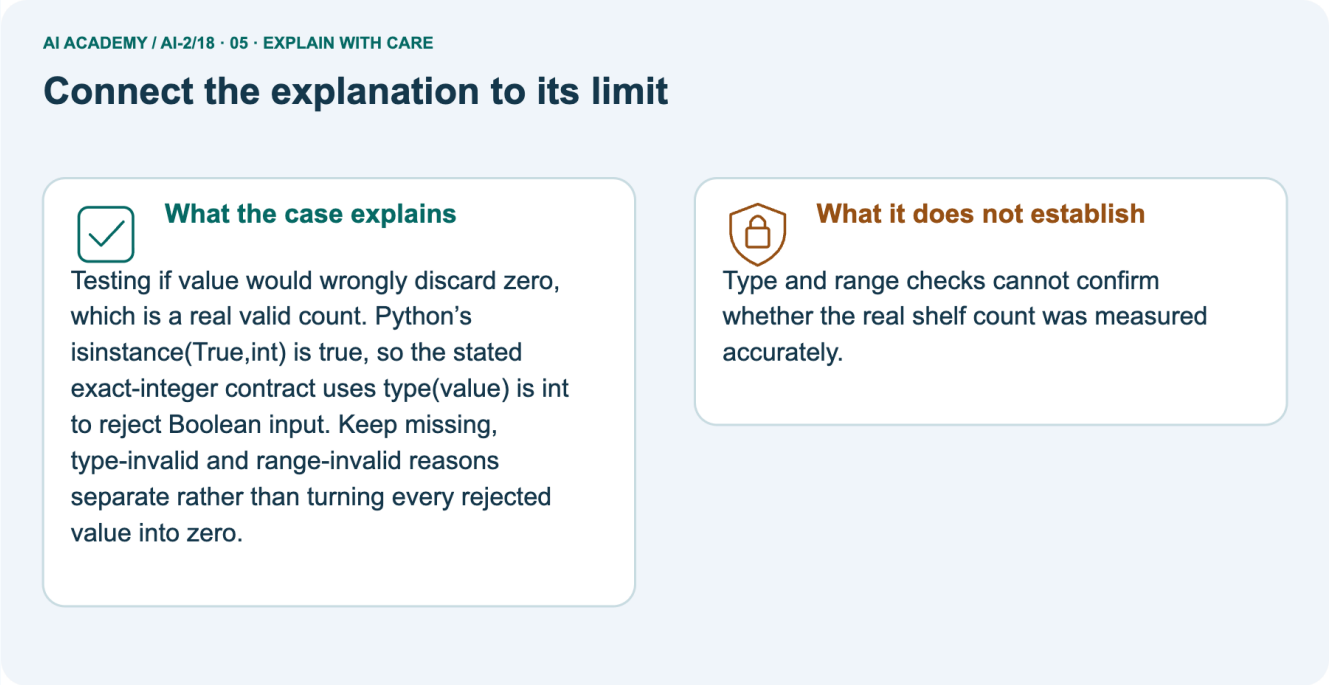
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The converted value now satisfies both type and range checks.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

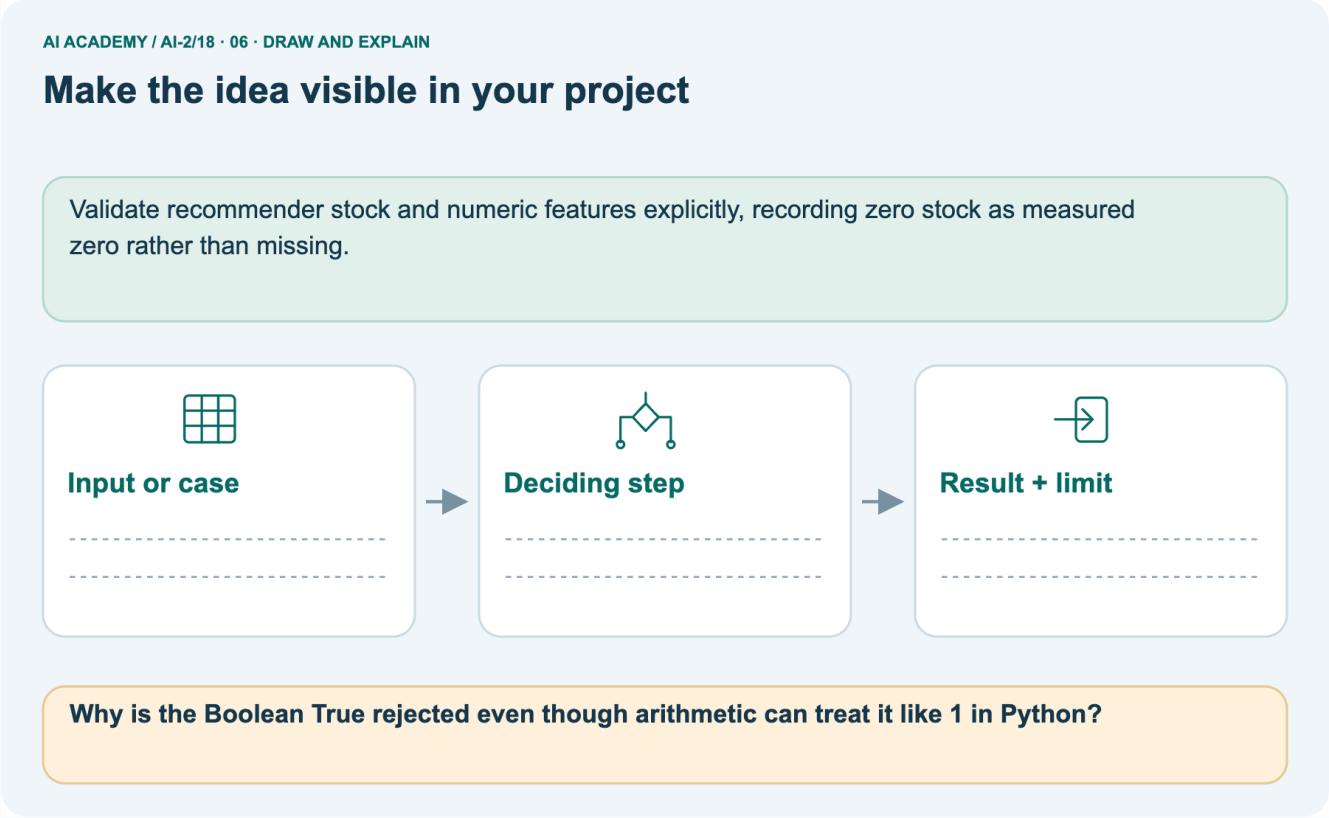
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Type and range checks cannot confirm whether the real shelf count was measured accurately.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The contract requires actual integer counts, not Boolean flags; type(True) is bool.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.